

# Map Services Running on Open Ports DEV-354

## Open Ports Services Mapping Documentation

Category: Network Security / Service Discovery

(Sanitized / Masked Version — Safe for Public Sharing)

**SECURITY NOTE:** All real IP addresses have been replaced with TARGET-01 to TARGET-15. This document contains no identifiable infrastructure information due to security measures.

### 1. Objective

The objective of this task was to identify which services are running on the TCP ports discovered during the open-port scanning phase. The process used Nmap service/version detection to map:

**Open Port → Service → Version/Technology (where identifiable)**

This provides a clearer view of the services exposed by the authorized target systems.

### 2. Service Mapping Approach

This reflects the actual project workflow:  
It follows the working pattern: target input → validation → full TCP scan → open-port extraction → service detection → result preservation.  
The actual service-detection command used was:

**nmap -Pn -sV -p "\$OPEN\_PORTS" "\$IP"**

The -sV option was used specifically to identify the service and version associated with previously discovered open ports.

### 3. Service Detection Configuration

| Category          | Configuration                        | Purpose                               |
|-------------------|--------------------------------------|---------------------------------------|
| Scanner           | Nmap 7.98                            | Service/version detection             |
| Targets           | 15 authorized systems (anonymized)   | Defined assessment scope              |
| Port Input        | Previously discovered open TCP ports | Focus service detection               |
| Service Detection | -sV                                  | Identify services and versions        |
| Host Discovery    | -Pn                                  | Scan without relying on ping response |
| Automation        | Bash                                 | Execute mapping across targets        |
| Concurrency       | Maximum 4                            | Controlled parallel execution         |

The project automation reads the target list, validates addresses, performs port discovery, extracts open ports, runs service detection, and stores the resulting data.

### 4. Service Mapping Results

The following table consolidates the service/version information identified during the completed scans. All real IP addresses have been replaced with anonymous TARGET identifiers.

| Target    | Port  | Service    | Detected Version / Technology    |
|-----------|-------|------------|----------------------------------|
| TARGET-01 | 22    | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17 |
| TARGET-01 | 54321 | MySQL      | MariaDB 5.5.5-10.6.23            |
| TARGET-02 | 22    | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.14 |
| TARGET-02 | 4646  | HTTP       | Golang net/http server           |
| TARGET-02 | 8500  | HTTP       | Golang net/http server           |
| TARGET-02 | 9000  | SSL/HTTP   | Golang net/http server           |
| TARGET-02 | 9001  | SSL/HTTP   | Golang net/http server           |
| TARGET-03 | 22    | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17 |
| TARGET-04 | 22    | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17 |
| TARGET-05 | 22    | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17 |
| TARGET-06 | 22    | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17 |
| TARGET-06 | 53    | TCP        | tcpwrapped                       |
| TARGET-06 | 80    | HTTP       | OpenResty web app server         |
| TARGET-06 | 443   | HTTPS      | OpenResty                        |
| TARGET-06 | 7212  | PostgreSQL | PostgreSQL DB 9.6.0 or later     |

Map Services Running on Open Ports DEV-354 (Sanitized)

|           |      |            |                                           |
|-----------|------|------------|-------------------------------------------|
| TARGET-06 | 7213 | Unknown    | Not identified                            |
| TARGET-06 | 9111 | Redis      | Redis key-value store                     |
| TARGET-06 | 9112 | HTTP       | Golang net/http server                    |
| TARGET-06 | 9180 | HTTP       | OpenResty web app server                  |
| TARGET-07 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-07 | 53   | TCP        | tcpwrapped                                |
| TARGET-07 | 80   | HTTP       | OpenResty web app server                  |
| TARGET-07 | 443  | HTTPS      | OpenResty                                 |
| TARGET-07 | 7212 | PostgreSQL | PostgreSQL DB 9.6.0 or later              |
| TARGET-07 | 7213 | Unknown    | Not identified                            |
| TARGET-07 | 9111 | Redis      | Redis key-value store                     |
| TARGET-07 | 9112 | HTTP       | Golang net/http server                    |
| TARGET-07 | 9180 | HTTP       | OpenResty web app server                  |
| TARGET-08 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-09 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-10 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-11 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-12 | 22   | SSH        | OpenSSH 9.6p1 Ubuntu 3ubuntu13.19         |
| TARGET-12 | 80   | HTTP       | nginx 1.31.2                              |
| TARGET-12 | 443  | HTTPS      | nginx 1.31.2                              |
| TARGET-12 | 8443 | HTTPS-alt  | Version not identified                    |
| TARGET-13 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-14 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-15 | 22   | SSH        | OpenSSH 8.9p1 Ubuntu 3ubuntu0.17          |
| TARGET-15 | 443  | SSL/HTTPS  | Service identified, version not confirmed |

The service detection output confirms, for example, MariaDB on port 54321, OpenResty/PostgreSQL/Redis services on TARGET-06, and nginx services on TARGET-12.

The remaining SSH mappings were also confirmed through the service-detection results for TARGET-05, TARGET-08, TARGET-09, TARGET-10, TARGET-11, TARGET-13, and TARGET-14.

## 5. Service Inventory Summary

The completed mapping identified the following major service categories:

| Service Category     | Examples Identified          |
|----------------------|------------------------------|
| Remote Access        | SSH / OpenSSH                |
| Web Services         | HTTP, HTTPS                  |
| Web Servers          | nginx, OpenResty             |
| Application Services | Golang net/http              |
| Databases            | MariaDB, PostgreSQL          |
| Data Store           | Redis                        |
| Protected/Unknown    | tcpwrapped, unknown services |

### Overall service picture

SSH → Most frequently detected service

Web → HTTP, HTTPS, nginx, OpenResty

Application → Golang net/http

Database → MariaDB, PostgreSQL

Data Store → Redis

Unidentified → Unknown, tcpwrapped

## 6. Non-Standard Port Mappings

The scan also identified services operating on ports that are not their commonly associated default ports.

Examples include:

- MariaDB → port 54321

- PostgreSQL → port 7212
- Redis → port 9111
- HTTP/Golang → ports 4646, 8500, 9000 and 9001
- HTTP/OpenResty → port 9180

This demonstrates why the project used full TCP port coverage rather than checking only commonly used ports. The full scan covers ports 1–65535.

## 7. Unknown and Uncertain Services

Not every open port could be mapped to a definitive service.

### TARGET-02

Nmap identified several ports where the service could not be conclusively determined:

- 53 — tcpwrapped
- 8301 — unrecognized
- 8502 — unrecognized
- 8503 — unrecognized
- 8600 — tcpwrapped

Other ports on the same host were successfully identified as SSH and Golang HTTP services.

### TARGET-06 and TARGET-07

Port 7213 remained unidentified on both systems.

Port 9112 returned HTTP data but Nmap did not fully classify the service. The response contained a MinIO-related server header, providing an indication of MinIO, but the scan did not establish this as a definitive service classification.

### TARGET-12

Port 8443 was detected as https-alt?, but the exact service/version was not confirmed.

### TARGET-15

Port 443 was detected as ssl/https?; Nmap did not provide a confirmed server/version identification.

Important: These results are intentionally retained as uncertain rather than assigning unsupported service names.

## 8. Service Exposure Analysis

The service mapping provides the following high-level observations:

### SSH Exposure

SSH on port 22 was the most consistently identified service across the target set. Most systems exposed OpenSSH 8.9p1, while TARGET-12 exposed OpenSSH 9.6p1.

### Web Service Exposure

Multiple systems exposed HTTP/HTTPS services through:

- OpenResty
- nginx
- Golang net/http

### Database Exposure

The scan identified database services exposed directly through non-standard ports:

- MariaDB on 54321
- PostgreSQL on 7212

### Data Store Exposure

Redis was identified on port 9111 on two targets.

### Application/Custom Services

Several high-numbered ports were associated with Golang HTTP services or remained unidentified, demonstrating that service exposure was not limited to conventional ports.

This section describes the discovered service surface only; it does not constitute a vulnerability assessment.

## 10. Key Findings

- 15 authorized systems were processed.
- Service mapping was performed using Nmap 7.98 with -sV.
- SSH/OpenSSH was the most frequently identified service.
- nginx and OpenResty were identified as web servers.
- Golang net/http services were detected on multiple non-standard ports.
- MariaDB was identified on port 54321.
- PostgreSQL was identified on port 7212.
- Redis was identified on port 9111.
- Several services operated on non-standard ports.
- Some ports remained unknown, tcpwrapped, or only partially identified.
- The raw service-detection evidence was preserved for verification.

## 14. Conclusion

The service-mapping phase successfully converted the open-port results into a structured service inventory across the 15 authorized targets.

The implementation identified common services such as SSH, HTTP and HTTPS, as well as database and application services including MariaDB, PostgreSQL, Redis, OpenResty, nginx, and Golang net/http. Where Nmap could not confidently identify a service, the result was preserved as unknown or uncertain rather than being inferred.

The final workflow can therefore be summarized as:

**Discover Ports → Extract Open Ports → Detect Services → Identify Versions → Preserve Evidence → Analyze Service Exposure**